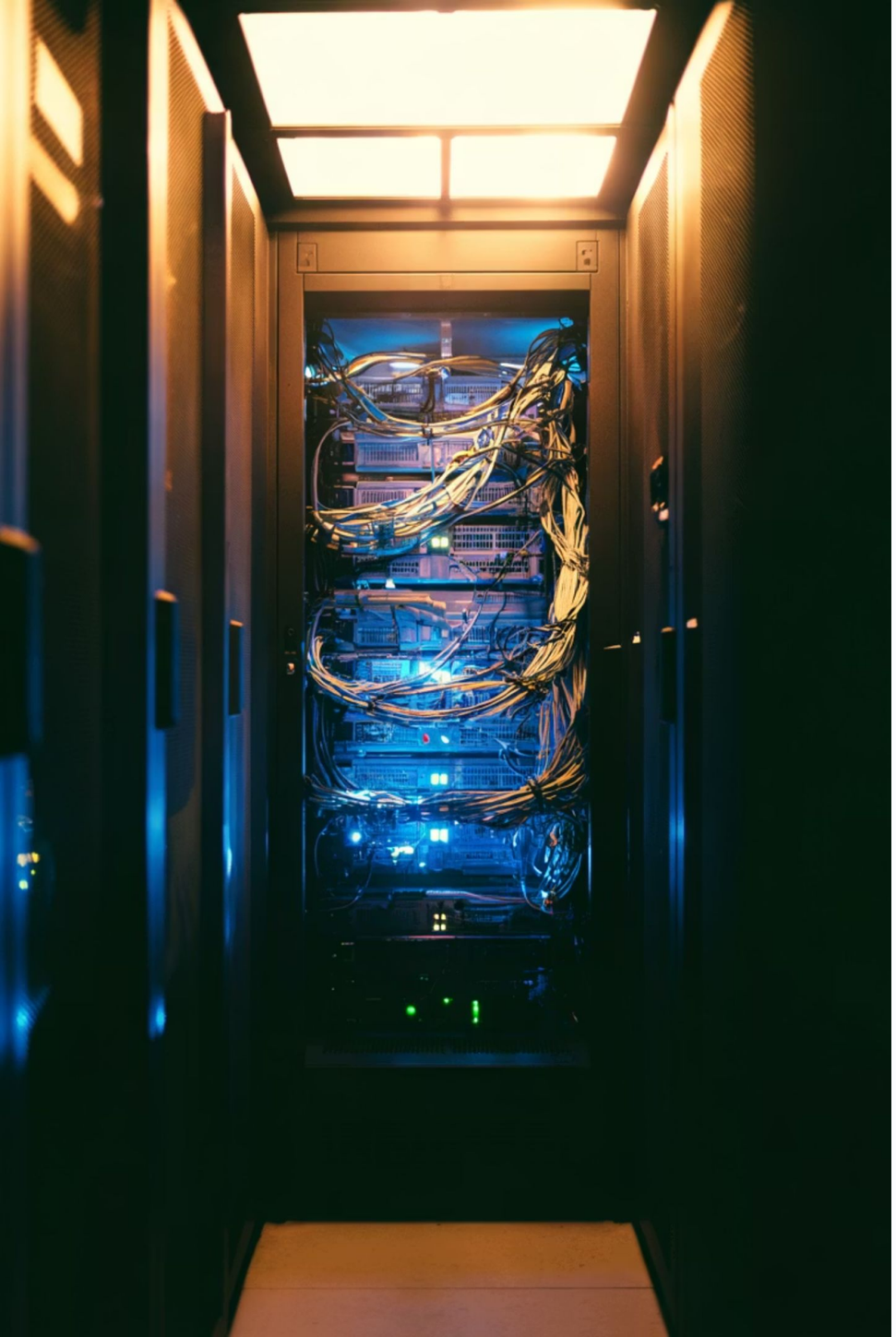

Threat Intelligence Aggregator

Automated OSINT Correlation & Blocklist Generation

APRIL 2026

Unified Mentor CYBERSECURITY INTERN

Harshada Rajale ·

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The Problem: Alert Fatigue & Manual Correlation

🔥 Overwhelming Volume

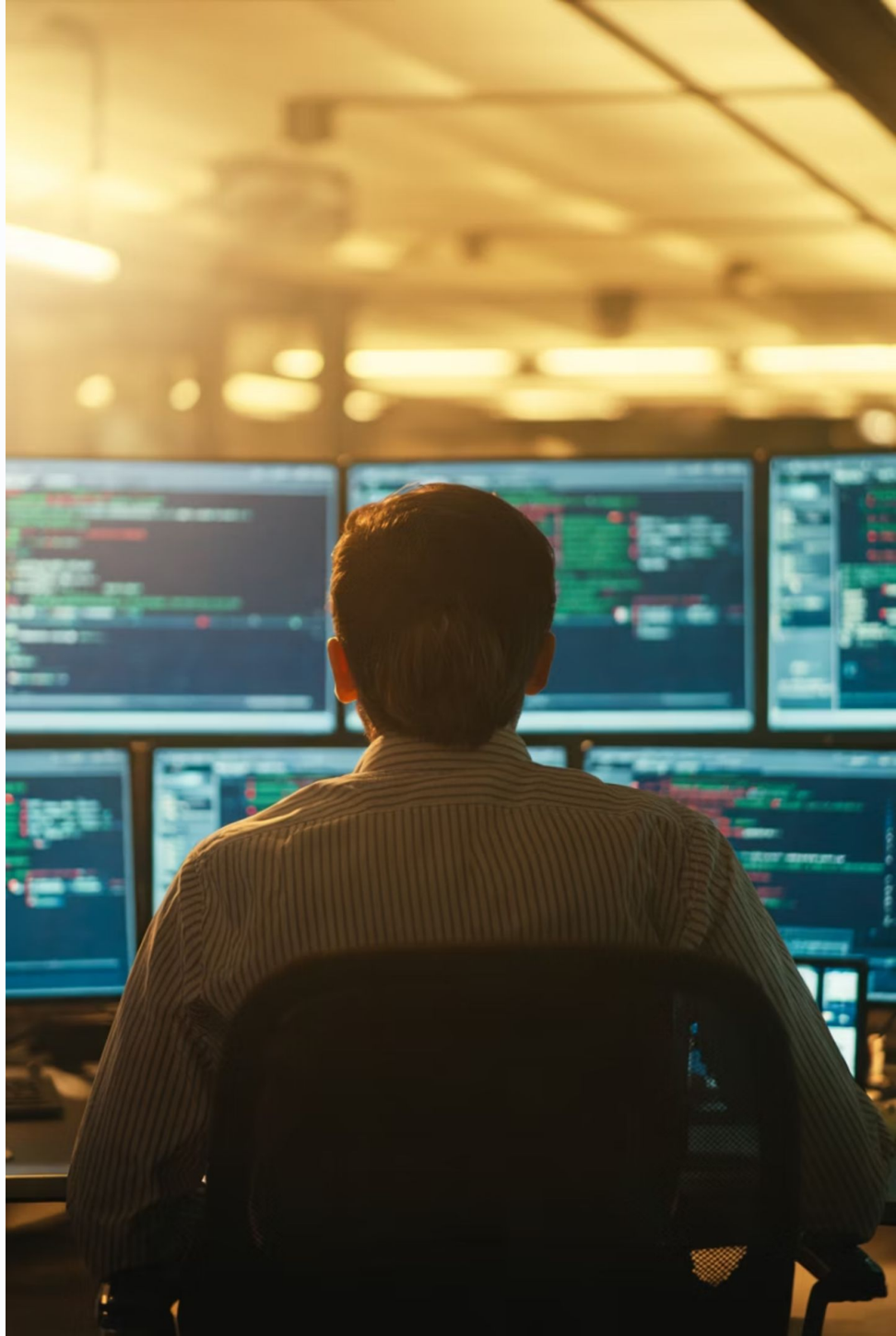
SOC analysts face massive daily influxes of OSINT threat feeds — far too many to triage manually without risking critical misses.

⚠️ Slow, Error-Prone Process

Cross-referencing external watchlists against thousands of internal firewall logs by hand is tedious and highly susceptible to human error.

🕐 Elevated MTTD

Every minute of delayed correlation raises Mean Time To Detect (MTTD), widening the window for an attacker to establish persistence.



Project Objectives

Automated Ingestion

Parse unstructured text watchlists (`.txt`) and structured firewall logs (`.csv`) without manual preprocessing.

Rapid Correlation

A fast, mathematically efficient engine identifies overlapping Indicators of Compromise (IOCs) across both data sources.

Actionable Output

Auto-generate a categorized `high_risk_blocklist.txt` file ready for direct deployment to network defense systems.

Interactive Sandbox

A live, client-side web application enables real-time demonstration and analyst testing without any server dependency.

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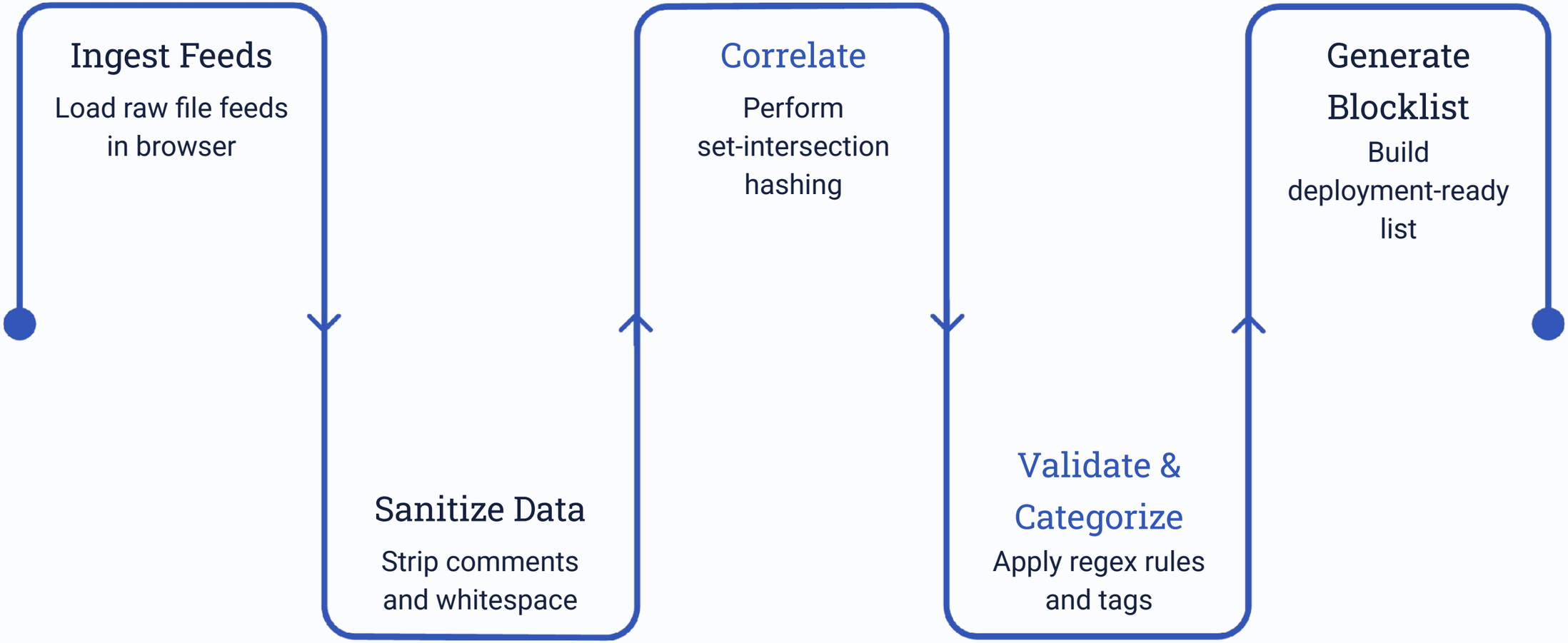
System Architecture & Data Flow



Each stage is executed client-side in the browser — raw feed ingestion flows through sanitization and hash-set correlation before Regex-driven categorization produces a deployment-ready blocklist with zero server round-trips.

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System Architecture & Data Flow

```
 [ Feed A: OSINT Watchlist (.txt) ]         [ Feed B: Firewall Logs (.csv) ]
                 |                                          |
                 v                                          v
[ Sanitize: Strip Comments & Whitespace ]    [ Extract: Isolate IOC Column ]
                 |                                          |
                 +---------------------+---------------------+
                                       |
                                       v
                   [ CORRELATION ENGINE (Set Intersection) ]
                   [   Cross-reference Feed A vs Feed B    ]
                                       |
                                       v
                            < Is there a match? >
                             /                 \
                        [ NO ]              [ YES ]
                          |                   |
                      (Discard)               v
                                  [ Validation & Categorization ]
                                  [  Regex: IP, Hash, or Domain ]
                                                |
                                                v
                                [ Output: high_risk_blocklist.txt ]
                                [    Trigger "HIGH RISK" Alert    ]
```

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Core Technologies & Algorithmic Efficiency

The Algorithmic Leap

The correlation engine converts arrays into JavaScript `Set` objects (hash sets), replacing a slow nested loop with a direct membership lookup.

Technology Stack

Frontend UI: HTML5 & CSS3 for responsive layout

Core Logic: Vanilla JavaScript (ES6+)

Data Structures: Native `Set` objects for O(1) lookup

Pattern Matching: Regex for IOC type classification (IP vs. Domain)

Before

nested loop — degrades severely at scale

After

Near-instant — processes thousands of logs without browser lag

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Operational Security (OpSec) Advantage

The Privacy Challenge

Corporate firewall logs contain highly sensitive internal IP addresses, connection histories, and network topology details that must never leave the organization's perimeter.

Client-Side Architecture = Zero Data Exposure

All JavaScript executes entirely within the analyst's local browser. **No firewall data is ever transmitted over the internet or uploaded to a third-party server.** Analysts can safely paste proprietary logs with zero risk of data leakage.

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Live Demo: Data Ingestion Interface

Dynamic Input Environment

The interface provides two independent input panels — **Feed A** accepts raw OSINT watchlists and **Feed B** accepts exported firewall logs.

- Paste or type raw text directly — no file upload required
- Modify feeds in real time between analysis runs
- Supports both IP addresses and domain-based IOC formats

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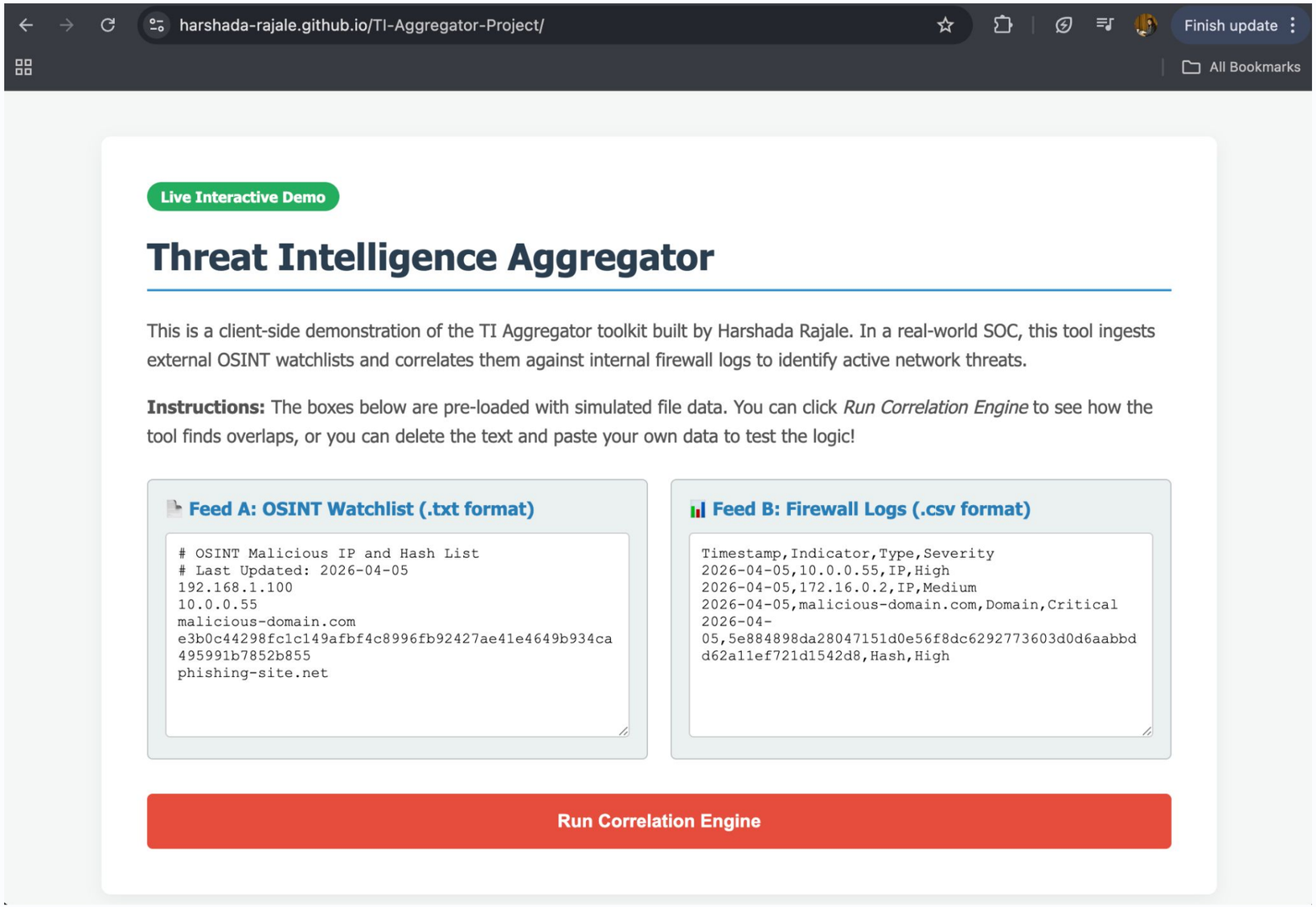
Live Demo: Data Ingestion Interface



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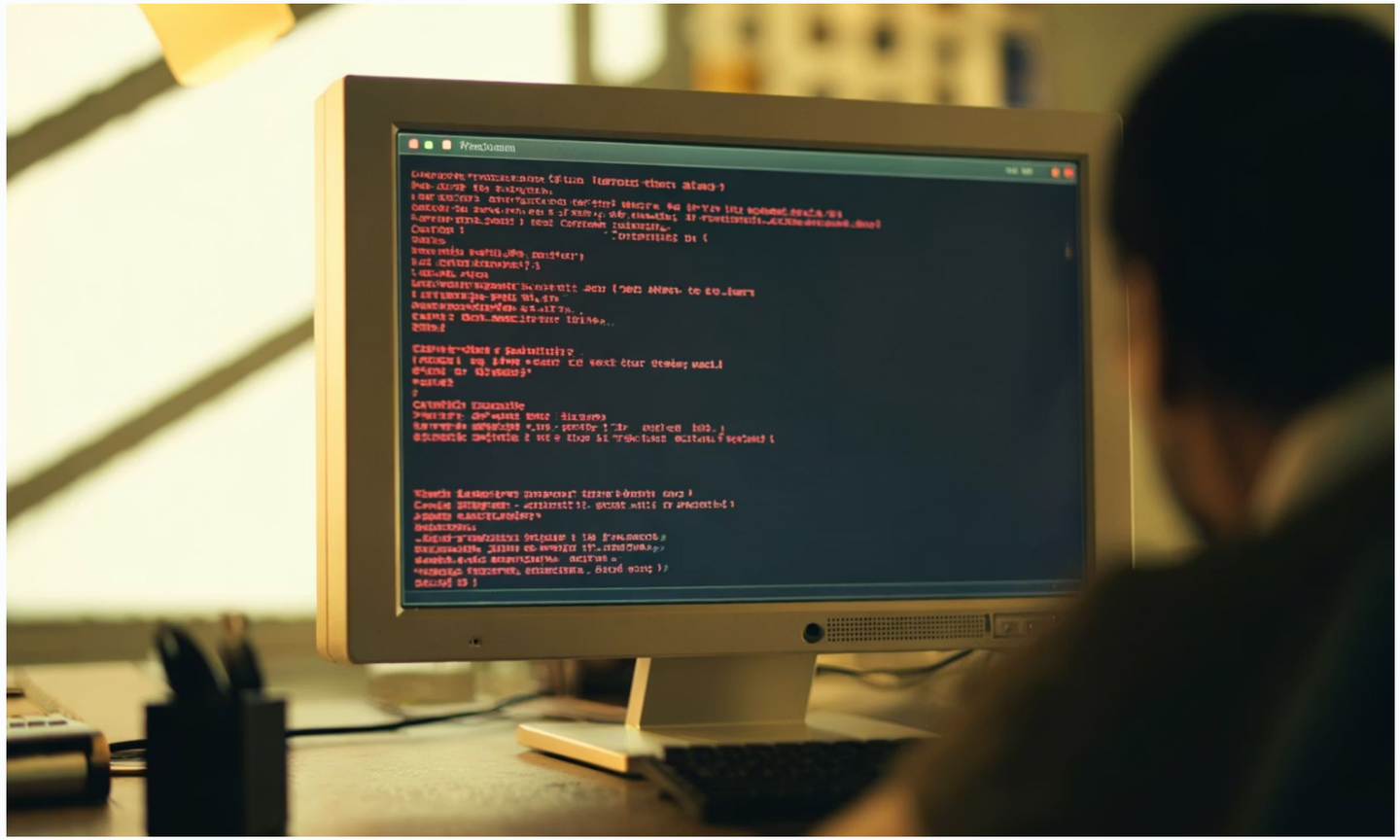
Live Demo: Correlation Engine Output



Engine Results at a Glance

The output panel surfaces every IOC that appears in both the threat watchlist and the firewall log, instantly classifying each match by type.

🔴 HIGH RISK

Matched IOCs flagged for immediate action

📋 Categorized

Regex distinguishes IP addresses from domain names

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Live Demo: Correlation Engine Output

harshada-rajale.github.io/TI-Aggregator-Project/

```
# OSINT Malicious IP and Hash List
# Last Updated: 2026-04-05
192.168.1.100
10.0.0.55
malicious-domain.com
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca
495991b7852b855
phishing-site.net
```

```
Timestamp,Indicator,Type,Severity
2026-04-05,10.0.0.55,IP,High
2026-04-05,172.16.0.2,IP,Medium
2026-04-05,malicious-domain.com,Domain,Critical
2026-04-
05,5e884898da28047151d0e56f8dc6292773603d0d6aabbd
d62a11ef721d1542d8,Hash,High
```

Run Correlation Engine

✅ **Analysis Complete! Blocklist Generated.**

Correlation Engine Output:

```
[!] HIGH RISK - Repeated IOC Found: 10.0.0.55 (IP Address)
[!] HIGH RISK - Repeated IOC Found: malicious-domain.com (Domain/URL)
```

[System Message] In a production environment, the high_risk_blocklist.txt file would be automatically exported and applied to the firewall.

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Conclusion & Future Enhancements

✅ What Was Achieved

Successfully designed, built, and deployed an automated, highly efficient, and privacy-preserving threat correlation tool that directly addresses real-world SOC analyst pain points — reducing manual effort and lowering MTTD.

🚀 Roadmap

Live API Integration

Pull real-time JSON feeds from AlienVault OTX and VirusTotal via REST APIs — eliminating manual paste-in workflows entirely.

SIEM Export Support

Format output as `.STIX` / `.TAXII` files for direct ingestion into Splunk, QRadar, and other enterprise SIEM platforms.

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Thank You & Q&A

Thank you for your time and review. Explore the live tool and source code via the links below.

🌐 Live Portfolio

Harshada-Rajale.github.io/TI-Aggregator-Project

💻 Source Code

github.com/Harshada-Rajale/TI-Aggregator-Project

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